

Darshan R

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EDUCATION

Krea University

2023 – 2026

- Bachelor of Science in Physics; CGPA: 7.79/10*

- Additional Concentration in Literature; Recipient of 100% tuition fee waiver
- Orientation Team '24, '25 — Physics Society — Literature Society — Politics Club

RESEARCH INTERESTS

Computational Neuroscience • Complex & Dynamical Systems • Physics of Neural Systems

EXPERIENCE

Computational Approaches to Memory and Plasticity

2026

- Indian Institute of Science, Education and Research, Pune*

- Attended a selective and intensive 15-day course on computational and theoretical neuroscience.
- Trained on various tools for modeling like NEURON, PyMoose, Brian2
- Worked on two team projects covering central pattern generators and network models.

Summer School on Dynamical Systems & Complexity

2025

- Society for Dynamical Systems, Indian Academies of Science & UPES, Dehradun*

- Selected in a cohort consisting primarily of postgraduate students.
- Studied synchronization, nonlinear dynamics and emergent behaviour.
- Presented *Modelling Brainwave Disorders Using Kuramoto Neural Networks*.

Teaching Assistant – PHYS110: Experiential Exploration of Physics

2025

- Krea University, Sri City*

- Guided laboratory demonstrations and discussions on foundational physics.

Chronos, The Physics Society

2024 – 2026

- Krea University, Sri City*

- Co-founded the university's first physics society, organizing and delivering the successful launch pitch to the physics faculty.
- Spearheaded organization's visual identity as Lead Designer, creating the official logo and social media templates.
- Designed interactive demonstrations of physical phenomena, including Lissajous figures, diffraction, and refraction, for the annual Physics Φ -esta festival (twice).

PROJECTS & PRESENTATIONS

Tritonia Swim Central Pattern Generator

- Adapted a reduced model of the CPG by Getting et al. (1989) from legacy NEURON/HOC into Python using Brian2.
- Led the computational modeling to analyze the rhythm's robustness against perturbations in input stimuli strength and maximum conductance.
- Conducted bifurcation analysis using XPPAUT to find eigenvalues, equilibrium points, and Hopf bifurcations.
- Co-presented findings at CAMP 2026

Zebra Finch Birdsong HVC Network Model

- Reproduced a neural network model of sequence generation by HVC neurons based on Jin et al. (2007).
- Used the reproduced model to investigate how intrinsic bursting and post-inhibitory rebound enhance network robustness to membrane noise.
- Co-presented findings at CAMP 2026

Modelling Brainwave Disorders Using Kuramoto Neural Networks

- Built Brian2 simulations of synchronization in coupled neural networks.
- Presented at the Indian Academies of Science Summer School on Dynamical Systems & Complexity.
- Extended the work to investigate neuronal avalanches, self-organized criticality and emergent dynamics.

History of Complex Numbers in Physics

- Presented on how complex numbers emerged in physics aimed at a general audience
- Connected the dots between history of science, physics and mathematics involved.

SKILLS

- **Computational Languages & Tools:** Python, XPPAUT, NEURON, JarDesigner, MCell
- **Packages & Data Standards:** Brian2, Pandas, PyMOOSE, NetworkX, DANDI, NWB
- **Soft Skills:** Scientific Communication & Teaching, Leadership, Creative Writing

AWARDS & HONORS

- Student Member, American Physical Society
- Student Member, Royal Aeronautical Society
- Affiliate Member, European Low-Gravity Research Association
- Winner, Epoch Delhi Hackathon (Hack Club)
- Gold Medalist of Mathematics, Class X
- Fellow, CatalystXL